



PREVALENCE OF ASTHMA DISEASE: IN CASE OF OBORRA PRIMARY
HOSPITAL OF OBORRA TOWN, HOROGUDURU WOLLEGA ZONE,
ETHIOPIA.

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ABSTRACT

Asthma disease is the most common respiratory disorder disease in developed and developing countries. The study was aimed to assess the factors that contribute for the cause of asthma disease. Questionnaires were the material used for this study and purposive sampling method was used for data collections. The study was conducted from sept 2023-Jul 2023. And it was conducted on a total of 74 sample size which was 40 males and 34 females. According to the study the more affected age group were 33-43(27%) and the major factor was air pollution(31%) this indicate the environment was polluted by different domestic wastes. the government should aware the society about the effect of asthma disease and the society should also the better mechanism of removing wastes.

The objective of this study was to assess the prevalence of asthma allergic disease among the patient visiting Oborra Hospital. To study such research secondary data is required. Data was collected from documented information. However, there were no enough documentary data. The data was analyzed by using table and bar graphs. The 54 adults in 2021 year, 186 in 2022 and 176 in 2023 were caused by asthma allergic disease. Among these three years higher rate of adults were infected in 2022 E.C. Year. Among all patients of the adults, 416 were caused by asthma allergic disease and other adults were free from asthma allergic disease. The other adults were suffering from disease, but they were negative of asthma as written on document of patients at Obora Hospital. Such problems of data were solved not only for conducted research but also for future study.

Keywords: Allergic, Asthma, Chronic, Episode, Inflammation and Impact.

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TABLE OF CONTENTS

CONTENOTES	PAGES
ACKNOWLEDGMENT.....	iii
CHAPTER ONE.....	- 1 -
1. INTRODUCTION.....	- 1 -
1.1. Background of the study.....	- 1 -
1.2. Statement of the problem.....	- 2 -
1.4. Significance of the study.....	- 3 -
1.5. Ethical considerations.....	- 3 -
1.6. Scope and Delimitation of the Study.....	- 3 -
1.7. Organization of the study.....	- 3 -
CHAPTER TWO.....	- 5 -
RELATED LITERATURE REVIEW.....	- 5 -
2.1. Concepts of Asthma Disease.....	- 5 -
CHAPTER THREE.....	- 7 -
3.1. Objectives of the study.....	- 7 -
3.2 General objective the study.....	- 7 -
CHAPTER FOUR.....	- 8 -
4.1. RESEARCH METHODOLOGY.....	- 8 -
4.2.. Description of the Study area.....	- 8 -
4.3. Study Design.....	- 8 -
4.4. Source of Data.....	- 8 -
4.5. Study Population.....	- 8 -
4.6. Sample size and sampling technique.....	- 8 -
4.7. Data collection tools and methods.....	- 9 -
4.8. Data Analysis.....	- 9 -

CHAPTER FIVE.....	- 10 -
5. RESEARCH RESULT AND DISCUSSION.....	- 10 -
5.1. Research Results.....	- 10 -
5.2. DISCUSSION.....	- 14 -
CHAPTER SIX.....	- 16 -
6.1 Conclusion and Recommendation.....	- 16 -
6.2. Conclusion.....	- 16 -
6.3. Recommendation.....	- 16 -
7. REFERENCE.....	- 17 -

TABLE CONTENTS

Table 1; - Socio-demographic characteristics of the study respondents.....	- 9 -
Table 2; - Age group s of the respondentes.....	- 10 -
Table 3;-The symptoms of asthma disease and number of respondents.....	- 11 -
Table 4; - Mechanisms of removing solid waste.....	- 12 -
Table 5; - Environmental contribution to cause asthma.....	- 12 -
Table 6; - The impact of asthma besides its health impact.....	- 12 -

CHAPTER ONE

1. INTRODUCTION

1.1. Background of the study

Asthma is chronic inflammatory disease of respiratory system (Global Initiative for asthma, 2009). The chronic inflammation is associated with airway hyper-responsiveness or an exaggerated airway narrowing response to, such as allergens and exercise, that lead to recurrent symptoms such as wheezing, dyspnea (shortness of breath), chest tightness and coughing. Several factors have been proposed to account for the cause of asthma disease and increase the prevalence (Jorge and Mazza, 2011). It is agreed that genetic changes in population would be too slow to account. Environmental factors have been a key role in the aetiology of asthma disease. Among the environmental determinants, exposure to allergens such as house dust mite, grass pollen, cockroach and animal dander (Bourdon *et al*, 2009).

Various methodologies have been used by researchers to measure adherence. These include the Medication Event Monitoring System (MEMS), patient self-report, patient pill-counts, pharmacy databases or refill rates, and blood level assessments. The MEMS uses a microprocessor technology that has been embedded in the cap of a container to record time and date the container with medication was opened (Aslani & Krass, 2009). Using the MEMS can generate incorrect results from patients who may remove more than the single recommended dose or may not remove a dose at all when they opened the bottle. (Aslani & Krass, 2009; Carter *et al*, 2010).

The global status report on non-communicable diseases (NCDs) indicates that NCDs kill more people annually than all other causes combined (WHO- GSR-NCDs, 2010). It noted that on the contrary to popular opinion 80% of NCD deaths occur in the low and middle income countries and that much of the human and social impact caused annually by NCD-related deaths could be averted through well-understood, cost-effective and feasible interventions (WHO- GSR-NCDs, 2010).

Investigator recently reported that in Canada, five thousand people die each year by this asthma disease (Kim and Mazza, 2011). And also asthma accounts for more absence from the school and more hospitalization of children and reproductive manpower at risk by this asthma disease. Moreover, in developing countries like Ethiopia, the effect of this disease is more serious.

(Yemaneberhan *et al*, 1997). Particularly, Oborra town is a place where several peoples are affected by asthma disease. Therefore; this study was assessing the factor that contributes for the cause of this disease. Asthma disparities have been attributed to differences in social economic status, but in African community when compared to Caucasian communities due to variety of social factors including disparities in health. More recent research has shown that other factors are more important in this inequality. There is a higher prevalence of poorer health outcomes of patients' asthma case assessments and communications, the social structure prescribed to patients (Bryant Stephens, 2009). Additionally, studies by Ricketts have shown that doctors (providers) prescribed the appropriate amount of medicine, at least the third of the patient would not take their medicines due to cultural and health belief (Bryant Stephens, 2009).

Haynes and colleagues (2008) reported that patients who are prescribed self-administered medications usually take less than half the prescribed doses (Haynes *et al*, 2008). Sub-optimal adherence to inhaled medicines for COPD and asthmatic conditions are reported by Mika and colleagues (2013), to be associated with poor symptom control and reductions in health-related quality of life, in addition to a higher healthcare utilization and cost (Mika *et al*, 2013).

Another reason why African communities have higher rates of adult asthma is because of residential segregation. One article illustrated that poor African communities that are separated from other ethnic communities have a higher concentration of poverty which leads to a lack of resources and support. According to a study in western Ethiopia, asthma accounted for 2% of outpatient and 5.4% of hospital admissions (Seyoum B, 1992).

As Ethiopia in general, Oborra town particularly, is a place where several peoples are affected by asthma disease. Therefore; this study was assessing the factor that contributes for the cause of this disease. Asthma disparities have been attributed to differences in social economic status, but in African community when compared to Caucasian communities due to variety of social factors including disparities in health. More recent research has shown that other factors are more important in this inequality.

1.2. Statement of the problem

Asthma is the most common respiratory disease in developing countries and developed countries. Asthma is a very difficult disease that is associated with other chronic respiratory diseases (Kim and Mazza, *al.*, 2011) and the symptoms of the asthma are also very similar with other

respiratory disorder disease like TB, Common cold and etc. Several factors have been proposed to account for the cause and its prevalence, even though environmental factors play a key role. Furthermore, asthma disease can affect all aged group of people all over the world (Kim and Mazza, 2011). Therefore, this study was attempted to answer the following questions: regarding to this study

1.3. Research questions

- ✓ What are the factors that contribute for the cause and effect of asthma disease?
- ✓ How the symptoms of asthma disease are identified from other respiratory disease?
- ✓ What are the main factor for the prevalence of asthma disease?

1.4. Significance of the study

Asthma is the worldwide problem affecting more than 230 million people around the globe. Therefore the finding of this study was provide enough information about the factors that aggravate asthma disease, Give awareness about the symptoms and effect of asthma disease, Be the base line for the further related study

1.5. Ethical considerations

Before the collection of data to be conducted, the purpose of a research study was clearly described to the organizations as well as each individual. All the data to be collected was corrected based on the willingness of the organizations to give information. The data also be kept confidential and only be used for the targeted research purposes. Data related to individual identification removed, and the study protocol was retrieved and approved by the advisers who to be assigned for the researcher.

1.6. Scope and Delimitation of the Study

This study presents the necessary components of newborn care practice definitions at an organizational level. The research results may be of an interest to academic studies related to the newborn care practice. A better understanding of newborn care practice characteristics on the public healthy success draws conclusions that can be beneficial not only for healthy science at profit field but also for other organizations and industries. At the same time, the researcher hope it be beneficial for healthy and policy makers for developing and finally improve their success.

1.7. Organization of the study

This paper was organize as follows. Chapter one the introduction, which includes background, statement of the problem, objectives, scope and Research question as well as the significance of

the study. Chapter two presents theoretical and literature reviews that deals with concepts and past studies and information pertinent to the study. Chapter three provides research methodology including description of the study area, sampling techniques, methods of data collection and tools for data analysis. This was follow by the results and analysis of different data source in chapter four. Finally, Chapter five was presents the conclusions and recommendation.

CHAPTER TWO

RELATED LITERATURE REVIEW

2.1. Concepts of Asthma Disease

Asthma is a chronic disease characterized by recurrent attacks of breathlessness and wheezing (World Health, 2000). Asthma remains the most common chronic respiratory disease (Ottawa and Ontario, 2007) affecting approximately 10% of the population. Although, asthma is often believed to be a disorder localized to the lungs, current evidence indicates that it may represent a component of symptoms of airway disease involving the entire respiratory tract, and this is supported by the fact that asthma frequently accompanies allergic rhinitis. Poor asthma control contributes to unnecessary morbidity, limitation to daily activities and impairment in overall quality of life (Bourdon *et al*, 2009). The chronic inflammation associated with airway hyperresponsiveness (an exaggerated airway narrowing response to triggers such as allergens and exercise), dyspnea (shortness of the breath), chest tightness and coughing.

A Swedish study also found increased prevalence of physician diagnosed asthma and an unchanged prevalence of most respiratory symptoms over an 11 year period that concluded that social awareness and changes in physician diagnostic practices may be the possible causes of increasing asthma reporting (Ekerljung *et al*, 2011).

In a systematic review of empirical studies in adults published between 1999 and 2009 conducted in France by Charles and colleagues (2011), to investigate the associations between perceptions of asthma and treatment beliefs and also adherence to inhaled corticosteroids. They noted that better adherence rates were associated with perception of the chronicity of asthma and its consequences on daily life, as well as the concept that it is necessary to continue treatment in the absence of symptoms. On the contrary, poor adherence rates were associated with the fear of side effects and the belief that treatment is ineffective in controlling symptoms (Charles *et al*, 2011).

Symptoms episodes are generally associated with appropriate treatment (Fitz Gerald *et al*, 2012). Asthma is associated with T-helper cell type 2 (Th2) immune response (Kim and Mazza, 2011) are typically of other atopic conditions various allergic such as dust mite, cockroach residue, furred animals, molds, pollen and non-allergic such as infections, tobacco smoke, cold air, exercise triggers produces a cascade of immune mediated events leading to chronic airway inflammations. Inflammation mediators such as histamine and cysteinyl leukotrienes that cause

bronchospasm (contraction of the smooth muscle in the air way), edema (swelling) and increase mucus secretion (Fitz Gerald *et al.*, 2012).

The mediators and cytokines released during an early phase of an immune response to an indicating allergen triggers a further inflammatory response(late phase asthmatic response)that lead to further air way inflammation and bronchial hyper reactivates(Lemansket *al.*,2011).The diagnosis of asthma should be suspected in patient with recurrent cough, wheeze, chest tightness and shortness of the breath such variable symptoms occur upon exposure to allergen or irritant, that worsen at night and that respond to appropriate asthma therapy are strongly suggestive of asthma Kaplan (*eta.*,2009). Alternative cause susceptible asthma symptoms should be excluded such as chronic obstructive pulmonary disease, bronchitis, chronic sinusitis gastro esophageal reflux disease, recurrent respiratory infection and heart disease.

A positive family history of asthma or other atopic disease and/or personal history of atopic disorder particularly allergic rhinitis, can also be helpful in identifying patient with asthma (Few *et al*, 2010), Exposure to agents encountered in the work environment can also cause asthma. The diagnosis of asthma in young children is often more difficult since episodic wheezing and cough are common and spirometry is a reliable in patients under 6years of age.

A useful method of confirming a diagnosis in young children is a trial treatment with short acting bronchodilators (Kim and Mazza, 2011). Spirometry is the preferred objective measure to assess for the reversible respiratory system obstruction (Yemanehberhan *et al*, 1997),(i.e. rapid improvement in lung function after elation of rapid acting bronchodilators and confirm a diagnosis of asthma).

During spirometer the patient is instructed to take the deepest breath possible and then to exhale as hard as fully as possible into the mouth piece the spirometry.Avoidance of relevant allergens or irritants is an important components of asthma management(Mengistu *al.*,2011) patients allergic to house dust mite should be instructed to use allergen impermeable covers for bedding and keep the relative humidity in the home below 50%(to inhibit mite growth).According to the recent estimates, asthma affects 300million people(Peter Costa,2007).Although people of all age suffer from the disease ,it most often starts childhood, especially children who have low birth weight,who are exposed to tobacco smoke and are raised in a low income environment. According to the latest data published asthma death in Ethiopia reached 6,743 or 1.12% of total death (World Health Organization, 2014).

CHAPTER THREE

3.1. Objectives of the study

3.2 General objective the study

To assess the health impact of asthma disease in Oborra hospital.

1.3.2 Specific objectives

1. To analyze the prevalence and effect of asthma disease in Oborra town.
2. To identify the symptoms of asthma disease are identified from other respiratory disease
3. To investigate which environmental factors are the main cause for asthma disease.

CHAPTER FOUR

4.1. RESEARCH METHODOLOGY

4.2.. Description of the Study area

The study was conducted in Oromia regional state, Horo Guduru Wollega, particularly Oborra town primary Hospital, Eastern Ethiopia. Oborra town is located at 400km away from Addis Ababa, which is the capital city of Ethiopia. All the soils types have good agricultural potentialities; annual rain fall varies between 1300mm and 1700mm respectively (CSA, 2013).

4.3. Study Design

The study was conducted on the patient's document of 2015 E.C in Oborra primary Hospital to determine the prevalence of asthma allergic disease among patients. Survey method of study design was used.

4.4. Source of Data

The source data was used secondary data which recorded in cards of patients who were visited at Oborra Hospital.

4.5. Study Population

Study population of research include all asthma diseases patients who was treated in oborra town and who going to visit in oborra primary hospital and Oborra town peoples who was assessable during the data collection.

4.6. Sample size and sampling technique

In this study the sample size was include all asthma diseases that register in log book patient recorded. Random sampling technique was appropriate to get the data; particularly purposive sampling technique was used on target individual to get information about typhoid fever disease. The sampling technique and sample size of the study was determined by using the following formula. Where, z = standard normal deviate at 95% level of confidence that is 1.96, $q=1-p=0.05$, d is margin of error 0.1. Therefore, the sample size calculated for residents is 74. Among the total population of oborra town 74 of them was selected as a sample. To distribute the questionnaires the person who come first was serve first

To distribute the questionnaires random sampling techniques and the sample for study (Martin, 1995). Sample size determination formula was be used.

Where, n =sample

d =margin error

z =confidence level is equal to 1.95

p =population proportion

$q=1-p=0.05\%$

$$n = \frac{Nz^2pq}{d^2 (N-1) + z^2 + pq}$$

$$n = \frac{2434(1.96)^2 + (0.05) / (0.05)^2(2434-1) + (1.96)^2(0.95) (0.05)}{1} = 74$$

$$n = 74$$

4.7. Data collection tools and methods

This research was conducted by using both primary and secondary sources of data. The primary data was collected using questionnaires which were closed ended questions and interviews. The secondary data was collected by introducing the objective of the study to obborra hospital. Then experts in the health institution was asked for giving a patients registration card and depending on the registered information of the year from 2022-2023, the patient s data was taken and the status level was known.

4.8. Data Analysis

The collected data was analyzed using data and other relevant statically tool. The results was present in the form of tables, graphs and pie chart. The data was analyzed by using tables and percent.

CHAPTER FIVE

5. RESEARCH RESULT AND DISCUSSION

5.1. Research Results

The study was conducted on a total of 74 of study sample size, 40 males and 34 females. The result of this study was obtained from the respondent of the study found in Oborra town. The data was surveyed through purposive sampling by questionnaire and they were questioned in the form of interview because the questionnaires were prepared in English, thus it must be translated in the language of the respondents (A/Oromo and Amharic). The result of the study were presented as follows.

Table 1; - Socio-demographic characteristics of the study respondents

Characters		Sex		Total	Percents(%)
		Males	Females		
Sex		40	34	74	100
Occupation	Students	6	9	15	20.27
	Employer	3	7	10	13.51
	Marchents	5	7	12	16.21
	Driver	5	0	5	6.75
	Total	40	34	74	100
Religion	Orthodox	7	9	16	21.62
	Muslim	5	4	14	18.78
	Catholic	7	5	12	16.21
	Protestantts	21	17	36	52.1
	Total	40	34	74	100
Ethnicity	Oromo	12	10	22	29.72
	Amhara	9	5	14	18.91
	Gurage	5	4	9	12.16
	Total	40	34	74	100
Family status	Single	18	15	33	44.59
	Married	22	19	41	55.4
	Total	40	34	74	100

Source: survey, 2023

In this study the more affected sex was male 40(40%) out of 74 total sample size of the study and the females were 34(34%). In the case of the occupation of the respondent, 10(13.5%) were labour force which indicated by Other in the table. The driver caused were only 5(6.75%) means the least. Regarding to the religion the highest affected were muslim 16(21.6%) whereas the least affected were protestant 11(14.86%), orthodox 25(33.78%), catholic 12(16.2%) and 10(13.5%) were others. Regarding to ethnicity 22(29.7%), 14(18.9%), 9(12%), and 29(39%) of the respondents were Oromo, Amhara, Gurage and Other respectively. In this case the highest affected were other which includes kefa, kulo, wolayta and the like. Whereas gurage were the least affected.

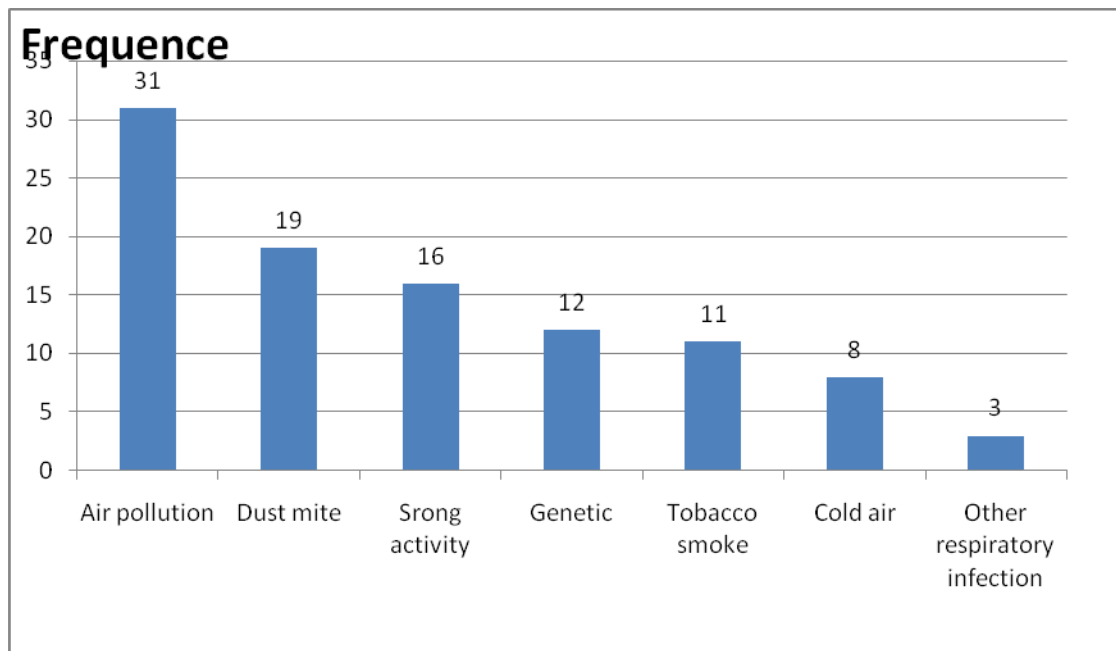
Table 2; - Age groups of the respondents.

Age groups	Sex		total	Percents(%)
	Males	Females		
1-10	2	1	3	4%
11-21	5	4	9	12.16%
22-32	10	7	17	22.97%
33-43	10	7	17	22.97%
44-54	5	6	11	14.86%
55-65	4	4	8	10.81%
66-76	3	3	6	8.10%
77-87	1	2	3	4%
Total	40	34	74	100

Source: survey, 2023

The above table indicates the age group from 33-43 years were the most affected 22(22.97%) and least affected was 77-87 years 3(4%).

Figure 1; - Factors that contribute for cuase of asthma disease.



Source: survey, 2023

The above figure clearly indicate which factors mostly contributed for the cause of asthma disease depending up on respondent given by study participant. Consequently air pollution was the major factor 31% of the total 100 study participant were caused by. Where as exercees 3% was the least factor.

Table 3;-The symptoms of asthma disease and number of respondents.

No	Symptoms	Frequency	Percent(%)
1	Cough or sneeze	20	20%
2	Shortness of breath	30	30%
3	Chest tight ness	25	25%
4	Wheeze	10	10%
5	Itching of nose	5	5%
6	Un able todo unusual work	15	15%

Source: survey, 2023

As the above table indicates shortness of breath was the stmtoms of most of asthmatic patients(30%) and itching of nose was (5%).

Table 4; - Mechanisms of removing solid waste.

Mechanism	Sex		Total	Parsent(%)
	Male	Female		
Burning	16	14	30	40.54%
Discarding on open field	13	11	24	32.43%
Accumulating and using as fertilizer	11	9	20	27
Total	40	34	74	100

Source: survey, 2023

The study was indicate the population of Obora town remove the domestic solid waste through the above mechanis

Table 5; - Environmental contribution to cause asthma.

Factors	Sex		Total	Parsent(%)
	Male	Female		
Working environment	29	19	48	64.86%
Domestic environment	11	15	26	35.1%
Toatal	40	34	74	100

Source: (own survey, 2023)

This indicate working environment was the major factor conrtibutar for the cause of asthma diseases that account 64.86% out of 74 respondant. And 35.1 % were caused by domestic environment.

Table 6; - The impact of asthma besides its health impact.

Impacts	Sex		Total	Parsent(%)
	Male	Female		
Social imact	13	11	24	32.4
Physichological imact	14	9	23	31
Economic impact	13	14	27	36.4
Total	40	34	74	100

Source: survey, 2023

As the above figure indicated asthma has other impacts besides its health impacts. Economic impact 37 % and psychological impact was 31 %.

5.2. DISCUSSION

In this study the more affected sex was male 40(40%) out of 74 total sample size of the study and the females were 34(34%). In the case of the occupation of the respondent, 10(13.5%) were labour force which indicated by Other in the table. The driver caused were only 5(6.75%) means the least. Regarding to the religion the highest affected were muslim 16(21.6%) whereas the least affected were protestant 11(14.86%), orthodox 25(33.78%), catholic 12(16.2%) and 10(13.5%) were others. Regarding to ethnicity 22(29.7%), 14(18.9%), 9(12%), and 29(39%) of the respondents were Oromo, Amhara, Gurage and Other respectively. In this case the highest affected were other which includes kefa, kulo, wolayta and the like. Whereas gurage were the least affected (own survey, 2019).

According to the recent study estimates asthma affects peoples of all age group (Peter costa, 2007) even though it most often starts at childhood. Similarly all age group of respondents were affected even though the most affected age group was (33-43) or 17(22.97%) out of total respondents of the study. The next most affected group was (44-54) which was 11(14.86%) out of total 100 respondents. Age group the least number of the respondent affected was (77-87) that was 3(4 %) (own survey, 2023)

Several factors have been proposed to account for the cause of asthma disease and increase the prevalence (Jorge and Mazza, 2011). Environmental pollution have been a key role in genesis of asthma disease (Bourdin *et al.*, 2009). Similarly this study shown the factors that contributes for the cause of asthma disease. Among the factors air pollution was mostly account to cause 31(31%), tobacco smoke 11(11%), dust mite 19(19%), cold air 12(12%), genetic factor 16(16%), other respiratory infection 8(8%) and exercise account only 3(3%). (Own survey, 2019).

The recurrent symptoms of asthma is wheezing, shortness of breath, chest tightness and coughing (Global Initiative for Asthma, 2009). In the same ways the symptoms of asthma of the respondents were cough, shortness of breath, chest tightness and wheezing commonly but some have itching nose and unable to do an usual work in addition to the first four symptoms. (own survey, 2023).

Those live domestic animals were affected by asthma because of animal danders ,hair and dust from animals were allergen for some individual. The habit of removing domestic solid waste was through different mechanisms, burning, discarding on open field and accumulating and using as fertilizers later. In this manner burning 30(40.5%), discard on open field 24(32.4%) and 20(27%) accumulate and use as fertilizers.

The working environment is more likely to be the cause of exposure related respiratory problems because of air quality poorer at work environment than domestic environment (Hendrick *et al.*, 2011). Similarly in this study working environment was shown to be the major environment to cause Asthma 48(64.86%) and the domestic environment 26(35.1%) to cause Asthma disease.

Besides of its health impact asthma has other different impact like social, economic, psychological and etc (Jorge and Mazza., 2011). In this study economic impact was shown to be the significant that 37(37%) were affected economically where as 32(32%) and 31(31%) were affected socially and psychologically respectively.

CHAPTER SIX

6.1 Conclusion and Recommendation

6.2. Conclusion

In general this study showed the factors that contribute for the cause of asthma disease in Obora hospitals. Those factors were tobacco smoke, air pollution, dust mite, cold air, genetic, exercise and other respiratory infection common cold, TB and the like. Among the factors the air pollution and dust mite were the most to account asthma disease. The environmental pollution of this area was believed to be resulted from improper removing of domestic wastes. To reduce this the society should be aware of the proper way of removing the wastes.

6.3. Recommendation

The government should be giving continuous awareness creation, the society about the effect of asthma, promoting stakeholder for fund raising, application previously done researches, and try to filling the gaps where the research are overviewed, and recommended, The society should properly remove the domestic waste. The health sector should work with the society towards prevention and control of asthma disease.

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Dear Respondents

This questionnaire is designed to help a study an Assessment of the Health Impact of Asthma disease (in case of primary Hospital of oborra town), n. This information is only used for academic purpose. So, I request you to answer all the questions below to the best of your information, in which please tick your choice in the box to show your response, and write down your own opinion, reason and suggestions in the space provided. Your information was handled with the highest order of confidentiality. Please don t write your name anywhere on the questions. I thank you for spending of your time to complete this research questionnaire.

I. General Information

1. Age -----

2. Work office-----

3. Sex----- A. Male B. Female

4. Educational Status

A. illiterate

B. 9-12Grade complete

C. 1-4 Grade Complete

D. Diploma

E. 5-8 Grade Complete

F. Degree and above

5. Work experience -----

6. Department/ section-----

Questionnaire: - for assessment on the factors that contribute for the cause of asthma disease in oborra town.

Part one: - personal information

Age _____

Job: A. Student ☐ B. Employer ☐ C .Merchant ☐ D .Driver ☐
E, another ☐

Religious: A. Muslim ☐ B.Orthodox ☐ C. Catholic ☐ D. Protestan ☐ E.
Other ☐

Ethnicity: A. Oromo ☐ B. Amhara ☐ C. Garage ☐ D. other ☐

Family status: A. Single ☐ B. Married ☐

Part two:-

1 Do you have any respiratory disorder disease?

A. yes ☐ B. NO ☐

2 If your answer for question number 1 is yes, what is it?

A .Common cold ☐ B. Allergy ☐ C. Asthma ☐ D. other. ☐

3. If your answer for question no 2 asthma, what do you think the factor that causes asthma?

A. tobacco smoke ☐ B. Environmental pollution ☐ C. Dust mite ☐
D. cold air ☐ E. exercise. ☐

4. At what age you have suffered to this asthma?

A. at child age ☐ Bat adult ☐ C. old age ☐

5. When it is difficult to you?

A .at night ☐ B, in the morning ☐ C.during the rain ☐ D. during warm
cctemperature. ☐

6. What are the symptoms of asthma disease?

7What your Environmental hygiene is look like?

A. clean ☐ B polluted ☐ C. somewhat clean ☐

8. If your answer for question no 7 is B what makes it or what is the reason behind?

A. Gass from burned waste ☐
mals in your house?

A. YES ☐ B. NO ☐

10. How do you remove the domestic solid waste from the environment?

A. By burning ☐ B. by simply discarding at far away from the environment ☐

C. By accumulating and using as fertilizers latter ☐

11. How do you remove the domestic liquid waste?

A. By piping to the river ☐ B. by accumulating in the whole and recycling ☐ C. By treating ☐

12. Which environment do you think is the main factor to cause asthma?

A. Net and clean ☐ B. Polluted by different unpleasant smell which have polluted by waste from the factor ☐ C. environment ☐

13. How the environment could be neat and clean?

14. Have you ever contact a doctor?

A. yes ☐ B. no ☐

15. If yes, what your doctor recommend you?

16. What kind of medication do you use to control your asthma?

A. oral tablet ☐ B. in haler oxygen ☐ C. liquid by syringe ☐ D. not exercising ☐

17. What are the impacts of asthma in your life beside its health impact?

A. Social impact B. Economic impact C. Psychological impact